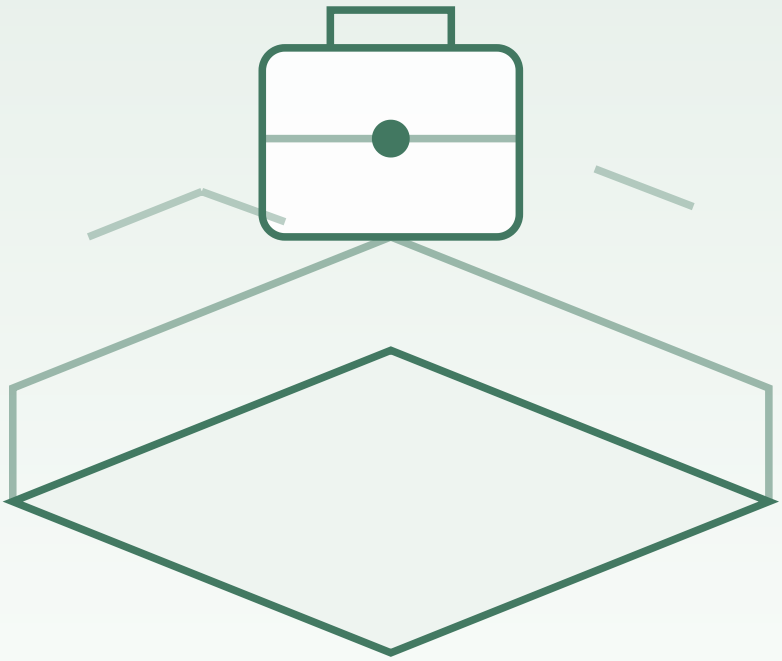
MYOFFICE · ENGINEERING

Portable Tools and Equipment System

What it must accomplish, how it works today, where it stands, and how to improve it.



DATE
12 October 2026

AUDIENCE
Owner, managers and developers

STATUS
Report only. No code or data was changed to write it.

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How to read this. Sections 1, 4, 5 and 6 are written for the owner and managers. Sections 2 and 3 explain the intent and the design. Section 7 and the appendices are for developers. Every figure is either measured from the code or the live database on 12 October 2026, or marked as an assumption or an owner decision.

1. Summary for the owner

The Tools system is meant to answer one question at any moment: *where is every portable tool, who may use it, and is it safe to use?* It does that for the Engineering register today, but the safety and approval half of the design is built and almost unused, and one whole part (gate passes) cannot be reached from the screen.

72
active tools (63 available, 2 out, 7 held for attention); 31 archived

754
eligibility records for 54 employees (who may use what)

240
movements and changes in history; 483 usage events

4
inspections recorded, against 72 tools that need regular checks

0
incidents, gate passes, evidence files and source registers stored

40
server endpoints; 33 server tests and 150 screen tests

Where it stands

- **Working and in use:** the register, issuing, returning, transferring and extending tools, the employee register, eligibility (trained, qualified, authorised), the history trail, notifications, Excel, Word and PDF downloads, and the Excel import.
- **Built on the server but barely used:** scheduled inspections, quarantine of failed tools, incident investigation, evidence photos, source-register storage. The data shows almost none of it has been exercised in real work.
- **Built on the server, no screen:** the gate-pass system (HOS, HOD, Security, Finance, General Manager sign-off with signed PDF). The screen for it was never connected and was removed in the 10 October rebuild.
- **Changed on 10 October:** the whole workspace was rebuilt on the shared MyOffice design. It passed every automated check, but only against test data. It has not yet been used on live data.

Top five improvements

1. **Rebuild the gate-pass screen** on top of the existing server code, so tools can legally leave site with a signed trail.
2. **Make inspections a habit, not a chore:** a due list per person, one-tap pass/fail, and a decision on the default check frequency for the 24 untagged hand tools.
3. **One sign-in.** Tools has its own accounts and passwords beside the main MyOffice login. Merge them so people have one identity and one place to be switched off.
4. **Prove the rebuild on live data** with a short, supervised walkthrough (issue, return, extend, mark ready), then retire the old behaviour notes.
5. **Close the paper gap:** barcode or QR labels on each tool, so issuing is a scan and not a search.

> **Decisions only the owner can make** are listed in section 6: the default inspection policy, the quarterly colour convention, who signs gate passes and when Finance is involved, and the authority for payroll recovery after negligence.

2. What the system must accomplish

The requirements come from the company's *Portable Tools and Equipment Management Procedure*, the management approval brief, and decisions recorded as the system was built (see the specification in `docs/TOOLS_SYSTEM_SPECIFICATION.md`).

2.1 Operational goals

#	Goal	Status today
1	One authoritative register for company and contractor tools and equipment	In use
2	Every item has a unique, structured register number (for example `PP-UG-ENG-TW-01`), allocated without duplicates	In use
3	Record ownership, technical identity, home storage, current location, storage conditions and maintenance needs	In use
4	Issue only to someone who is trained and qualified and authorised, with none of the three expired	In use
5	Track custody: issue, transfer, return, extend; due dates; overdue and six-hour escalation; full history	In use
6	Schedule and evidence pre-use, weekly, monthly, quarterly, calibration, maintenance, storage-audit and repair checks	Built, little used
7	Quarantine failed or unsafe tools until an authorised person records why they are fit again	Built, 7 tools held
8	Record loss, damage, theft, missing parts and late returns with an investigation and approved recovery terms	Built, unused
9	Control movement off site with a signed gate pass: HOS, HOD, Security, optional Finance, General Manager	No screen
10	Keep original source registers and photos without replacing the structured register	Built, unused

2.2 Who can do what

Role	Read	Register and compliance	Move tools	Manage accounts
Viewer	Registers they are allowed to see	No	No	No
Issuer	Own department	Own department	Own department	No
Administrator	Everything	All departments	No, unless also given an Issuer account	Yes

Gate-pass signing roles (HOS, HOD, Security, Finance, General Manager) are separate from these three roles. Signing needs the person to confirm their password or a numeric PIN again.

2.3 Business rules the system must enforce

- A contractor-owned item must name the contractor.
- A required pre-use check must be confirmed before issue.

- An item out on loan has one current custodian; each move is a permanent history event.
- Overdue is worked out from due dates, with a separate alert after six hours late, without a timer.
- A failed inspection puts the tool on hold. Releasing it needs a written resolution.
- Repair versus replace is judged against the procedure threshold (repair at or under 60% of the price of a new one).
- Gate-pass approvals go in order. A rejection ends the process.
- Every change must be repeat-safe, version-checked and reversible.

2.4 Qualities it must have

- **Truthful screens:** a failed request must never look like an empty register or a zero.
- **Resilient loading:** independent sources load separately so one slow source cannot block the register.
- **Audit trail:** every operational change records who, when and what, plus before-and-after snapshots.
- **Works on a phone and a desktop**, by keyboard, at larger text sizes.
- **No secrets in the browser:** service credentials stay on the server.

2.5 What stays outside the software

Quality of physical inspections, delivery of training, supervisors' competence, security checks, locks, CCTV, GPS and physical labels remain human and site responsibilities. The software records and enforces the process; it cannot do these things.

3. How it works today

3.1 Architecture

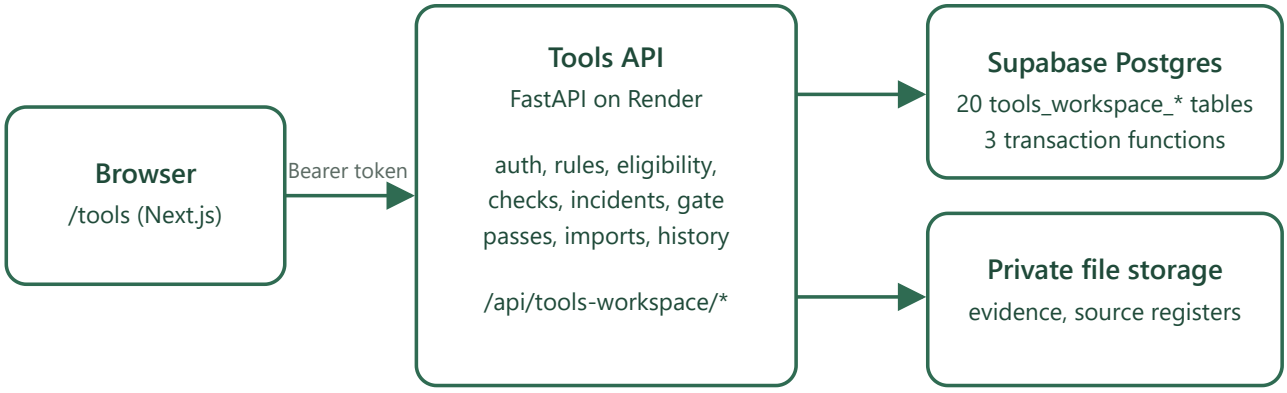


Figure 1. System context. The Tools system has its own accounts and sessions, separate from the main MyOffice sign-in.

3.2 Data model

Twenty tables, all prefixed `tools_workspace_`, created by seven migrations. Row-level security is enabled on the compliance tables; the server uses a service credential and never exposes it to the browser.

Group	Tables	Purpose
People and access	accounts, sessions, employees, competencies	Who can sign in, who can borrow what
Register	equipment, number_sequences, evidence, source_registers	The tools, their numbers, photos and original registers
Custody and audit	history, changes, idempotency	Every movement, reversible snapshots, repeat protection
Compliance	inspections, incidents	Checks and investigations
Gate passes	gate_passes, gate_pass_items, gate_pass_approvals	Sequential approval to move tools off site
Product insight	usage, errors, feedback, notification_reads	How it is used and what breaks

3.3 Equipment states

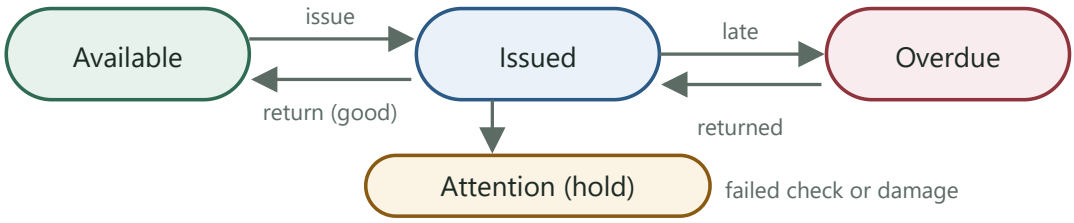


Figure 2. Simplified equipment life cycle. A held tool returns to Available only through a written “mark ready” resolution. Archived tools sit outside the cycle.

3.4 The screens (rebuilt 10 October 2026)

Section	What a person does there
Overview	See counts, what needs attention, and the latest movements; jump straight to a filtered list.
Equipment	Browse tools as cards or a table, filter by status, category and location, open one tool in a side panel, and issue, receive, transfer, extend, edit, archive or mark ready. Select several to archive together.
In use	The same register limited to tools currently out, overdue first.
Employees	By person or by equipment: who may use what, grant or remove eligibility with a named authoriser, deactivate people.
Compliance	Overdue inspections, open incidents, expiring approvals, and forms to record checks, report incidents and close investigations.
History	The permanent issue and return trail, searchable.
Source registers	Upload and keep original PDF, spreadsheet or scan registers.
Account access, Analytics, Feedback	Administrators only: roles and gate-pass signing rights, usage trends, suggestions with audio.

Shared with the rest of MyOffice since 10 October: the frame, sidebar, top bar, dialogs, forms, tables and charts. The Tools-only stylesheet (about 139 KB) and its private controls were deleted.

> **Behaviour changes in the rebuild you should confirm (see also section 6):** equipment pictograms and the quick Issue/Receive button on cards were removed (open the side panel instead); free-text fields use the browser's own suggestion list (no Tab-to-complete); the "Explore" pop-ups on analytics are gone; typeface and text size now follow the app-wide setting.

3.5 Security and audit

- Passwords and PINs are verified by the server. The browser keeps only a session token.
- Every change goes through one transaction function that checks the version, refuses repeats (idempotency key), and writes the equipment record, the history line and a reversible snapshot together.
- Department scope is enforced on the server for Issuer and Viewer accounts.
- Notification read state is per account. Usage and error capture never blocks the register.

4. Where it stands

4.1 Live snapshot (12 October 2026, read-only count)

Item	Count	What it tells us
Active tools	72	63 available, 2 out, 7 held for attention. 31 more are archived.
Employees	54	All active; about 16 are demonstration records to switch off.
Accounts	3	Very few people actually sign in to Tools.
Eligibility records	754	The main piece of real data entry; imported and maintained.
History events	240	Real use, but small: roughly three per active tool.
Inspections	4	Far below what the check rules imply. See section 5, gap 2.
Incidents, gate passes, evidence files, source registers	0 each	These features are built but have never been used.
Feedback, usage events	5, 483	Used, but little feedback sent.

4.2 Size and tests

757
lines in the server router, 40 endpoints, 20 tables, 7 migrations

33
server tests (eligibility, numbering, checks, gate passes, imports)

150
screen tests in 20 files, passing on 11 October; 44 source files; no accessibility violations found by the automated scan

4.3 What is verified and what is not

Area	Evidence	Confidence
Server rules and database functions	Server test suite; production migrations applied and checked in the September audit	High
Register, issue, return, eligibility on live data	In daily use; bulk archive confirmed live on 7 October; Excel import run live on 8 October	High
The 10 October rebuilt screens	Test data only: unit tests, a browser walkthrough of every screen, a real issue request captured, accessibility scans	Medium
Gate-pass flow end to end	Server code and tests only. No screen. Never run with real people	None in practice
Phones and the installed app	Narrow-screen emulation only, no real device	Medium
Evidence photo upload, source-register upload	Server tests; never used live	Medium

5. Gaps and risks

#	Gap	Why it matters and the evidence	Severity
1	**No gate-pass screen**	The server can create a pass, collect the signed sequence with password or PIN, and produce a signed PDF, but no screen reaches it. Tools can leave site without the signed trail the procedure requires. Zero passes exist.	High
2	**Inspections barely recorded**	Only 4 inspections exist for 72 tools that default to monthly and quarterly checks. Either checks are not being done, or they are not being recorded; the system cannot tell which. Either way the “overdue check” blocks and warnings are probably noisy and trusted by no one.	High
3	**Two sign-ins**	Tools has its own accounts and passwords (3 accounts), separate from MyOffice. Leavers must be switched off in two places; people keep two passwords; there is no single audit identity.	Medium
4	**Rebuild not yet proven live**	The new screens passed every automated check on test data. Rare paths (transfer with gate-pass references, extend, mark ready, evidence upload) have not been exercised on real records.	Medium
5	**Policy decisions open**	The quarterly colour mapping is provisional. Whether Finance joins every pass or only above a value is undecided. Authority for payroll recovery after negligence is unconfirmed. The default check frequency for the 24 hand tools added without tags is unconfirmed.	Medium
6	**Demo and incomplete data**	About 16 demonstration employees remain active. Purchase date and cost are blank for imported tools, so value-based rules (CCTV above a value, repair threshold) have nothing to work with.	Medium
7	**Lifecycle holes**	Gate passes cannot be cancelled or closed. Incidents jump from open to closed with no “investigating” step, so overdue investigations are hard to manage.	Low
8	**Manual identification**	Tools are found by searching names or numbers. Physical labels exist, but there is no scan. Searching costs time at the tool crib and invites mistakes.	Low
9	**No calibration certificate store**	Calibration is a date and an outcome, but the certificate itself is not stored or linked, which an audit under a quality system would ask for.	Low
10	**Thin management reporting**	Downloads exist, but no view of utilisation, tools never used, repeat damage by person or department, overdue patterns, or cost of loss.	Low
11	**Single point of failure on the server's free tier**	The API sleeps when idle and wakes slowly. The screens retry for this, but a cold start at the start of a shift delays the first issue.	Low

Risks if nothing changes

- **Compliance risk:** without gate passes and recorded inspections, the procedure is on paper and not in the system.
- **Trust risk:** constant warnings about unrecorded checks teach people to ignore all warnings, including real ones.
- **Loss risk:** a tool that leaves site has no signed record, and the investigation tools are untested when a real loss happens.

6. Improvement roadmap

Ordered by value and by what unblocks what. Effort is a rough size for one developer: S under a day, M a few days, L one to two weeks.

Now (next two weeks)

Improvement	What it delivers	Effort	Needs from owner
Prove the rebuild on live data	A supervised walkthrough: issue, return, extend, transfer, mark ready, upload a photo. Fix what the walkthrough finds. Confirm or reverse the behaviour changes in section 4.3.	S	An hour with a real Issuer
Inspection policy and a due list	Decide default frequencies per tool class. One list "my checks due this week" with one-tap pass/fail. Fewer, truer warnings.	M	Check frequencies per class
Clean the data	Switch off the demo employees, fill in purchase date and cost where known, review the 7 held tools.	S	Records and a decision

Next (one to two months)

Improvement	What it delivers	Effort	Needs from owner
Gate-pass screen	Create a pass from selected tools, show the sign-off chain, sign with password or PIN, download the signed PDF, add cancel and close. Built on the existing server code.	L	Who signs; the Finance rule
Single sign-in	Sign in to Tools with the MyOffice account. Roles and approval rights move to a table linked to the main user. One place to switch someone off.	L	A migration go-ahead
Incident workflow	Add the “investigating” step, due-date reminders, and link incidents to the person and the tool in the side panel.	M	Recovery authority
Labels and scanning	Print QR labels from the register; scan to open a tool or to issue and receive. Works with a phone camera, no new hardware.	M	Label size and stock

Later (a quarter)

Improvement	What it delivers	Effort	Needs from owner
Calibration certificates	Store the certificate with the calibration record; remind before expiry; block issue of out-of-calibration instruments.	M	Supplier list
Management reports	Utilisation, tools never used, repeat damage, overdue patterns, cost of loss, per department. Scheduled email.	M	Which questions matter

Offline-friendly issuing	Queue issue and return on a phone when the signal drops underground and sync later, with the same repeat protection the server already has.	L	Real signal map
Link to Maintenance and Spares	A failed tool opens a work order; parts used appear against the tool; cost of ownership per tool.	L	Process agreement

Decisions only the owner can make

1. Default inspection frequency for tagged and untagged hand tools, and which classes need none.
2. The official quarterly colour convention (currently a provisional mapping).
3. Gate-pass signatories, and whether Finance signs every pass, only above a value, or only for contractor equipment.
4. Who may confirm negligence and approve payroll recovery, and what evidence is required.
5. Whether to keep or restore the removed card pictograms and the quick Issue/Receive button.

Good-practice references for the owner

These are general references for framing the improvements, not company policy: calibration and measuring-equipment control (ISO 9001 clause on monitoring and measuring resources; ISO 10012), asset management principles (ISO 55000 family), and, for lost or damaged items, the recovery and fair-process principles of the applicable labour law. They should be checked against the company's own procedure before adoption.

7. Developer guide

7.1 Where things live

Path	What it is
`backend/app/routers/tools_workspace.py`	All Tools endpoints, request models and rules (eligibility, check due dates, status, gate-pass credential check)
`backend/supabase_migration_tools_workspace*.sql`	Seven additive migrations; the three transaction functions are created or replaced here
`backend/tests/test_tools_workspace_router.py`	Server tests, 33 cases
`frontend/app/tools/useToolsWorkspace.ts`	All state, data loading and every action. No layout code. Start here to change behaviour
`frontend/app/tools/workspaceLoad.ts`	Progressive, shape-checked loading of the independent sources
`frontend/app/tools/ToolsFrame.tsx`	Frame: sidebar, top bar, search, notifications, settings, profile
`frontend/app/tools/ToolsSections.tsx`, `ToolsSectionBar.tsx`, `RegisterView.tsx`	The open section, its figures and toolbar, and the register as cards or table
`frontend/app/tools/ToolDrawer.tsx`, `ToolsModals.tsx`, `ToolsForms.tsx`	Tool side panel, the one dialog for all forms, and the forms themselves
`frontend/app/tools/ToolsPeople.tsx`, `ToolsEligibility.tsx`, `ToolsCompliance*.tsx`, `ToolsAuth.tsx`, `ToolsAnalytics.tsx`	Employee and eligibility screens, compliance, accounts, analytics
`frontend/app/tools/toolSelectors.ts`, `toolEligibility.ts`, `prototype.ts`	Pure selection, counting and eligibility logic, and the domain types. Keep it free of React
`frontend/components/shared/floor/`	The app-wide look: backdrop, animated text, sound, theme
`frontend/scripts/route-specs/tools.mjs`	Browser route check for `/tools` with fixture data

7.2 Adding a rule or a field, end to end

1. Write the rule in `docs/TOOLS_SYSTEM_SPECIFICATION.md` first: the clause, who, the trigger, the decision, the evidence.
2. If it needs storage, add an *additive* migration file. Never edit an applied one. Re-create the transaction functions if the field is part of an equipment change.
3. Add or extend the Pydantic model and the rule in `tools_workspace.py`. Write the server test first.
4. Map the field in `workspaceLoad.ts` (`ServerTool`), `toolFromServer.ts` and `prototype.ts` (`Tool`), then the form in `ToolsForms.tsx` and the record in `ToolDrawer.tsx`.
5. Make the screen truthful: loading, empty, failed and not-allowed states, never an empty list for an error.
6. Run the checks below, and record what was and was not verified.

7.3 Commands

Purpose	Command (from the folder shown)
Server tests	`backend/.venv/Scripts/python.exe -m pytest tests/test_tools_workspace_router.py` (use a short `--basetemp` on Windows)
Typecheck and lint	`frontend: npx tsc --noEmit` and `npx eslint app/tools`
Screen tests	`frontend: npx vitest run app/tools`
Browser route check	`frontend: MSYS_NO_PATHCONV=1 node scripts/verify-routes.mjs --only /tools`
Documentation ledger	`frontend: npm run docs:ledger` then `npm run docs:check`

7.4 Conventions and traps

- Use the shared UI components (`@/components/ui-system`). Do not add a Tools-only stylesheet or private controls.
- Dates through `lib/format.ts` . Status colours through the shared status tones, never hand-picked.
- The Tools sign-in is not the MyOffice sign-in: its session lives in browser storage under `myoffice.tools.session.v2` with a browser key. Test harnesses set both.
- The server's database client sorts with `order(col, desc=...)` ; the `asc=` form raises an error that test fakes can hide. Use a strict fake in tests.
- Do not run migrations, push, or change live data without the owner's explicit go-ahead for that batch.
- Imports: the Excel and Word importers preview first and skip existing register numbers. Never apply an import without a preview and a backup of the rows it will change.

8. Appendices

A. Server endpoints (prefix `/api/tools-workspace`)

Area	Endpoints
Authentication and accounts	POST /auth/register, POST /auth/login, GET /auth/me, GET /accounts, PATCH /accounts/{id}, PUT /auth/signing-pin
Notifications	GET /notifications, POST /notifications/read
Employees	GET /employees, POST /employees, PATCH /employees/{id}
Equipment	GET /tools, POST /tools, PATCH /tools/{id}, POST /tools/{id}/archive, POST /tools/{id}/mark-ready, POST /tools/{id}/evidence
Custody	POST /tools/{id}/commands (issue, return, transfer, extend), POST /tools/{id}/issue, POST /tools/{id}/return, GET /history
Compliance	GET /compliance, POST /competencies, POST /tools/{id}/inspections, POST /tools/{id}/incidents, POST /incidents/{id}/close
Gate passes	GET /gate-passes, POST /gate-passes, POST /gate-passes/{id}/decision, GET /gate-passes/{id}/pdf
Source registers and imports	GET and POST /source-registers, POST /imports/preview, POST /imports/commit
Change control	GET /changes, POST /changes/{direction} (restore or redo a snapshot)
Insight	POST /analytics/usage, POST /analytics/errors, GET /analytics, POST /feedback

B. Database tables (20)

accounts, sessions, employees, equipment, history, changes, idempotency, evidence, source_registers, number_sequences, competencies, inspections, incidents, gate_passes, gate_pass_items, gate_pass_approvals, notification_reads, usage, errors, feedback. Transaction functions: next register number, apply change, restore change.

C. Glossary

Register number	The structured unique ID of a tool: site, area, department, activity, sequence.
Eligible	Trained, qualified and authorised, with none expired, for that tool or its category.
Held / Attention	A tool that failed a check or was recorded as faulty and cannot be issued until marked ready.
Gate pass	A signed approval to move tools off site or between areas.
Idempotency key	A repeat-protection token: sending the same change twice has the effect of once.
HOS, HOD	Head of Section, Head of Department.

D. How the numbers in this report were produced (12 October 2026)

- Endpoint, table and migration counts: counted from `tools_workspace.py` and the seven migration files.
- Test counts: 33 functions in the server test file; 150 passing screen tests under `app/tools` on 11 October.
- Live counts: a read-only count query on the production database on 12 October. No record was read or changed beyond counts.

- Earlier facts (the 8 October Excel import, bulk archive on 7 October, demo employees): the project's dated notes.
- Not verified for this report: behaviour on a real phone, the gate-pass flow, evidence upload on live data.

E. Source documents in the repository

`docs/TOOLS_SYSTEM_SPECIFICATION.md`, `docs/TOOLS_WORKSPACE.md`, `docs/TOOLS_DESIGN_STANDARD.md`, `docs/DALLAGLIO_AUDIT_2026-09-26.md`, `docs/MUSE_HANDOFF_2026-10-10.md`, `components/ui-system/README.md`.